

Unit 2 Advanced Math

2.1 Equivalent Expressions

1

★ Mark for Review

ABC

The equation $(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$ is true for all x , where a and b are constants. What is the value of ab ?

A. 18

B. 20

C. 24

D. 40

2

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ABC

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

I. $x - 9$ II. $3x - 7$

A. I only

B. II only

C. I and II

D. Neither I nor II

3

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ABC

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x , what is the value of $\frac{a}{b}$?

4

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ABC

The equation $\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$ is true for all $x > 2$, where r and t are positive constants. What is the value of rt ?

A. -20

B. 15

C. 20

D. 60

5

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ABC

For what value of x is the given expression $\sqrt[5]{70n} \left(\sqrt[6]{70n} \right)^2$ equivalent to $(70n)^{30x}$, where $n > 1$?

6

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ABC

The expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a, b, c are constants. What is the value of b ?

7

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ABC

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

A. $\frac{1}{(x+1)(4x-5)}$

B. $\frac{3}{3x-6}$

C. $-\frac{1}{(x+1)(4x-5)}$

D. $\frac{9}{(x+1)(4x-5)}$

8

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ABC

In the expression $\frac{x^2 - c}{x - b}$, b and c are positive integers. If the expression is equivalent to $x + b$ and $x \neq b$, which of the following could be the value of c ?

A. 4

B. 6

C. 8

D. 10

9

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ABC

The expression $0.36x^2 + 0.63x + 1.17$ can be rewritten as $a(4x^2 + 7x + 13)$, where a is a constant. What is the value of a ?

10

★ Mark for Review

ABC

In the expression $3(2x^2 + px + 8) - 16x(p + 4)$, p is a constant. This expression is equivalent to $6x^2 - 155x + 24$. What is the value of p ?

A. -3

B. 7

C. 13

D. 155

11

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ABC

The expression $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following?

A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$ B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$ C. $\frac{y\sqrt{y}}{x\sqrt{x}}$ D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

12

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ABC

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $(hx + k)(x + j)$, where h, k , and j are integers. Which of the following must be an integer?

A. $\frac{b}{h}$

B. $\frac{b}{k}$

C. $\frac{45}{h}$

D. $\frac{45}{k}$

13

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ABC

Let $f(x) = x^3 - 9x$, $g(x) = x^2 - 2x - 3$. Which of the following expressions is $\frac{f(x)}{g(x)}$ equivalent to, for $x > 3$?

A. $\frac{1}{x+1}$

B. $\frac{x+3}{x+1}$

C. $\frac{x(x-3)}{x+1}$

D. $\frac{x(x+3)}{x+1}$

14

★ Mark for Review

ABC

The expression $\frac{1}{3}x^2 - 2$ can be rewritten as $\frac{1}{3}(x - k)(x + k)$, where k is a positive constant. What is the value of k ?

A. 2

B. 6

C. $\sqrt{2}$ D. $\sqrt{6}$ **15**

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ABC

Which of the following is equivalent to $\left(a + \frac{b}{2}\right)^2$?

A. $a^2 + \frac{b^2}{2}$ B. $\frac{a^2 + b^2}{4}$ C. $\frac{a^2 + ab}{2} + \frac{b^2}{2}$ D. $a^2 + ab + \frac{b^2}{4}$

16

★ Mark for Review

ABC

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$?

A. $\frac{xy+y+4}{x^2y-16x+64xy}$

B. $\frac{xy+9+12}{x^2y-8xy+x-8}$

C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$

D. $\frac{xy^2+13xy-8y}{x^2y-16x^2y+64xy}$

17

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ABC

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

18

★ Mark for Review

ABC

The equation $\frac{x^2+6x-7}{x+7} = ax+d$ is true for all $x \neq -7$, where a and d are integers. What is the value of $a+d$?

A. -6

B. -1

C. 0

D. 1

19

★ Mark for Review

ABC

Which of the following expressions is equivalent to $\frac{x^2 - 2x - 5}{x - 3}$?

A. $x - 5 - \frac{20}{x - 3}$

B. $x - 5 - \frac{10}{x - 3}$

C. $x + 1 - \frac{8}{x - 3}$

D. $x + 1 - \frac{2}{x - 3}$

20

★ Mark for Review

ABC

The expression $(7532 + 100y^2) + 10(10y^2 - 110)$ can be written in the form $ay^2 + b$, where a and b are constants. What is the value of $a + b$?

21

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ABC

The expression $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$ is equivalent to ax^b , where a and b are positive constants and $x > 1$. What is the value of $a + b$?

22

★ Mark for Review

ABC

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

A. $(x + a)^2$

B. $(x - a)^2$

C. $(x + a)(x - a)$

D. $x^2 - ax - a^2$

23

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ABC

If a and c are positive numbers, which of the following is equivalent to $\sqrt{(a + c)^3} \cdot \sqrt{a + c}$?

A. $a + c$

B. $a^2 + c^2$

C. $a^2 + 2ac + c^2$

D. a^2c^2

2.2 Non Linear and Systems of Equations

1

★ Mark for Review

ABC

What is the positive solution to the equation $2|4 - x| + 3|4 - x| = 25$?

2

★ Mark for Review

ABC

One solution to the equation $x^2 - 2x - 9 = 0$ can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

A. 8

B. 10

C. 20

D. 40

3

★ Mark for Review

ABC

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

4

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ABC

Given the system:

$$\begin{cases} x - y = 1 \\ x + y = x^2 - 3 \end{cases}$$

Which ordered pair is a solution?

A. $(1 + \sqrt{3}, \sqrt{3})$

B. $(\sqrt{3}, -\sqrt{3})$

C. $(1 + \sqrt{5}, \sqrt{5})$

D. $(\sqrt{5}, -1 + \sqrt{5})$

5

★ Mark for Review

ABC

If x is a solution to the equation $\frac{1}{x^2 + 10x + 25} = 4$, which of the following is a possible value of $x + 5$?

A. $\frac{1}{2}$

B. $\frac{5}{2}$

C. $\frac{9}{2}$

D. $\frac{11}{2}$

6

★ Mark for Review

ABC

During a 5-second time interval, the average acceleration a (m/s²) of an object with an initial velocity of 12 m/s is defined by the equation

$$a = \frac{v_f - 12}{5}$$

If this is rewritten as $v_f = xa + y$, what is the value of x ?

7

★ Mark for Review

ABC

Which of the following is a solution to $2x^2 - 2 = 2x + 3$?

A. 2

B. $1 - \sqrt{11}$ C. $\frac{1}{2} + \sqrt{11}$ D. $\frac{1 + \sqrt{11}}{2}$ **8**

★ Mark for Review

ABC

In the xy -plane, a line with equation $2y = c$ intersects a parabola $y = -2x^2 + 9x$ at exactly one point. What is the value of c ?

9

★ Mark for Review

ABC

What is the smallest solution to the equation

$$\sqrt{(x-2)^2} = \sqrt{3x+34}$$

10

★ Mark for Review

ABC

Given the equation

$$57x^2 + (57b + a)x + ab = 0$$

where a and b are positive constants. The product of the solutions is kab . What is the value of k ?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57

11

★ Mark for Review

ABC

In the equation $-x^2 + bx - 676 = 0$, b is a positive integer and the equation has no real solution. What is the greatest possible value of b ?

12

★ Mark for Review

ABC

If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?

13

★ Mark for Review

ABC

In the equation $2x^2 - 4x = t$, t is a constant. If the equation has no real solutions, which of the following could be the value of t ?

A. -3

B. -1

C. 1

D. 3

14

★ Mark for Review

ABC

In the equation $x(kx - 56) = -16$, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

15

★ Mark for Review

ABC

The equation $\frac{14x}{7y} = 2\sqrt{w+19}$ relates the distinct positive real numbers w , x , and y . Which equation correctly expresses w in terms of x and y ?

A. $\sqrt{\frac{x}{y}} - 19$

B. $\sqrt{\frac{28x}{14y}} - 19$

C. $\left(\frac{x}{y}\right)^2 - 19$

D. $\left(\frac{28x}{14y}\right)^2 - 19$

16

★ Mark for Review

ABC

Solve for x : $\sqrt{2x+6} + 4 = x + 3$ What is the solution set of the equation above?

A. $\{-1\}$

B. $\{5\}$

C. $\{-1, 5\}$

D. $\{0, -1, 5\}$

17

★ Mark for Review

ABC

The function f , defined by

$$f(x) = -\frac{1}{2}(x - 4)^2 + 10$$

If the function g (not shown) is defined by $g(x) = -x + 10$, what is one possible value of a such that $f(a) = g(a)$?

18

★ Mark for Review

ABC

In the equation $x^2 - 34x + c = 0$, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

19

★ Mark for Review

ABC

In the equation $-16x^2 - 8x + c = 0$, c is a constant. The equation has exactly one solution. What is the value of c ?

20

★ Mark for Review

ABC

How many distinct real solutions does the equation $(x - 1)^2 = -4$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

21

★ Mark for Review

ABC

How many distinct real solutions does the equation $5x^2 + 10x + 16 = 0$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

22

★ Mark for Review

ABC

Given the system:

$$\begin{cases} y = x^2 + 2x + 1 \\ x + y + 1 = 0 \end{cases}$$

If (x_1, y_1) and (x_2, y_2) are the two solutions to the system, what is the value of $y_1 + y_2$?

A. -3

B. -2

C. -1

D. 1

23

★ Mark for Review

ABC

If $u - 3 = \frac{6}{t - 2}$, what is t in terms of u ?A. $\frac{1}{u}$ B. $\frac{2u + 9}{u}$ C. $\frac{1}{u - 3}$ D. $\frac{2u}{u - 3}$

24

★ Mark for Review

ABC

In the equation $-9x^2 + 30x + c = 0$, c is a constant. The equation has exactly one solution. What is the value of c ?

A. 3

B. 0

C. -25

D. -53

25

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ABC

In the equation

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39,$$

where c is a positive constant, which of the following is one of the solutions?

A. $-c$ B. $-c^2 - 39^2$ C. $-\sqrt{39^2 - c^2}$ D. $-\sqrt{c^2 + 39^2}$

26

★ Mark for Review

ABC

The graphs of the equations

$$\begin{cases} 8x + y = -11 \\ 2x^2 = y + 341 \end{cases}$$

intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

A. -15

B. -11

C. 2

D. 8

27

★ Mark for Review

ABC

What is the sum of the solutions to the equation

$$x(x + 1) - 56 = 4x(x - 7)?$$

28

★ Mark for Review

ABC

In the xy -plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$, where a and b are constants. What is the slope of line p ?

A. 6

B. 2

C. -2

D. -6

29

★ Mark for Review

ABC

The formula

$$D = T - \frac{9}{25}(100 - H)$$

approximates the dew point D in Fahrenheit, given temperature T in Fahrenheit, and relative humidity $H\%$, where $H > 50$. Which of the following expresses H in terms of D and T ?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

30

★ Mark for Review

ABC

Given the equation

$$x - 29 = (x - a)(x - 29),$$

where a is a constant and $a > 30$, which of the following are solutions?

I. a

II. $a + 1$

III. 29

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

31

★ Mark for Review

ABC

Given the system:

$$\begin{cases} y = 2x^2 - 21x + 64 \\ y = 3x + a \end{cases}$$

In the system of the equations, a is a constant. The graphs intersect at exactly one point (x, y) . What is the value of x ?

A. -8

B. -6

C. 6

D. 8

32

★ Mark for Review

ABC

In the xy -plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a) . What is the value of a ?

33

★ Mark for Review

ABC

Given the equation

$$64x^2 - (16a + 4b)x + ab = 0$$

The sum of the solutions is $k(4a + b)$. What is the value of k ?

34

★ Mark for Review

ABC

What is the sum of the solutions to the equation

$$x^2 - 40x - 10 = 0$$

A. 0

B. 5

C. 10

D. 40

35

★ Mark for Review

ABC

Given the system:

$$\begin{cases} y = x^2 + 3x - 7 \\ y - 5x + 8 = 0 \end{cases}$$

How many solutions are there?

A. Exactly 4

B. Exactly 2

C. Exactly 1

D. None

36

★ Mark for Review

ABC

Solve for x :

$$\frac{4x^2}{x^2 - 9} - \frac{2x}{x + 3} = \frac{1}{x - 3}$$

A. -3

B. $-\frac{1}{2}$ C. $\frac{1}{2}$

D. 3

37

★ Mark for Review

ABC

Given the equation

$$64x^2 + bx + 25 = 0$$

For which of the following values of b will the equation have more than one real solution?

A. -91

B. -80

C. 5

D. 40

38

★ Mark for Review

ABC

Given the system:

$$\begin{cases} y = -1.5 \\ y = x^2 + 8x + a \end{cases}$$

If the system has exactly one real solution, what is the value of a ?

39

★ Mark for Review

ABC

In the xy -plane, the graph of $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

A. $-\frac{481}{4}$

B. -100

C. $-\frac{319}{4}$

D. $-\frac{9}{2}$

40

★ Mark for Review

ABC

Given the system:

$$\begin{cases} x^2 + y + 7 = 7 \\ 20x + 100 - y = 0 \end{cases}$$

The solution is (x, y) . What is the value of x ?

2.3 Non Linear Function

1

★ Mark for Review

ABC

The quadratic function h is defined as $h(x) = 2(x - 4)^2 - 32$. In the xy -plane, the graph of $y = h(x)$ intersects the x -axis at the points $(0, 0)$ and $(t, 0)$, where t is a constant. What is the value of t ?

A. 1

B. 2

C. 4

D. 8

2

★ Mark for Review

ABC

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

A. $(0, 14)$ B. $(0, 2)$ C. $(0, 22)$ D. $(0, -8)$

3

★ Mark for Review

ABC

The function f is defined by $f(x) = 9,000(0.66)^x$. The function models the number of advertisements a company sent to its clients each year, where x is the number of years since 1997, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the xy -plane, which of the following is the best interpretation of the y -intercept?

A. The minimum estimated number of advertisements the company sent to its clients during the 5 years was 1,708.

B. The minimum estimated number of advertisements the company sent to its clients during the 5 years was 9,000.

C. The estimated number of advertisements the company sent to its clients in 1997 was 1,708.

D. The estimated number of advertisements the company sent to its clients in 1997 was 9,000.

4

★ Mark for Review

ABC

The first term of a sequence is 9. Each term after the first is 4 times the preceding term. If w represents the n th term of the sequence, which equation gives w in terms of n ?

A. $w = 4(9^n)$

B. $w = 4(9^{n-1})$

C. $w = 9(4^n)$

D. $w = 9(4^{n-1})$

5

★ Mark for Review

ABC

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

6

★ Mark for Review

ABC

A machine launches a softball from ground level. The softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds and hits the ground at 3.6 seconds. Which equation represents the height above ground h , in meters, of the softball t seconds after it is launched?

A. $h = -t^2 + 3.6$

B. $h = -t^2 + 51.84$

C. $h = -16(t - 1.8)^2 - 3.6$

D. $h = -16(t - 1.8)^2 + 51.84$

7

★ Mark for Review

ABC

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2, 0)$ and a y -intercept at $(0, -323)$. What is the value of b ?

8

★ Mark for Review

ABC

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function $f(x) = -500x^2 + 25,000x$, where x is the unit price of the product in dollars. The graph of $y = f(x)$ intersects the x -axis at 0 and a . What does a represent?

- A. The revenue, in dollars, when the unit price of the product is \$0
- B. The unit price, in dollars, of the product that will result in maximum revenue
- C. The unit price, in dollars, of the product that will result in a revenue of \$0
- D. The maximum revenue, in dollars, that the company can make

9

★ Mark for Review

ABC**Growth of a Culture of Bacteria**

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8 ?

- A. Day 5
- B. Day 9
- C. Day 11
- D. Day 12

10

★ Mark for Review

ABC

The function $f(x) = ax^2 + 4x + c$ is a quadratic function, where a and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that opens upward and has a vertex at point (h, k) . If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

I. $c < 0$ II. $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II

11

★ Mark for Review

ABC

The function $f(x) = 4x^2 + 64x + 262$ is defined, and $g(x) = f(x + 5)$. For what value of x does $g(x)$ reach its minimum?

A. -13 B. -8 C. -5 D. -3

12

★ Mark for Review

ABC

The function $f(x) = (x - 6)(x - 2)(x + 6)$ is defined. In the xy -plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

13

★ Mark for Review

ABC

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of 7 meters?

A. 3

B. 6

C. 7

D. 9

14

★ Mark for Review

ABC

Given $y = x^2 - 14x + 22$, for what value of x does the value of y reach its minimum?

15

★ Mark for Review

ABC

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 15$ has a y -intercept at $\left(0, -\frac{99}{7}\right)$. The product of a and b is $\frac{65}{7}$. What is the value of a ?

16

★ Mark for Review

ABC

The function $f(x) = (x - 14)(x + 19)$ is defined. For what value of x does $f(x)$ reach its minimum?

A. -266

B. -19

C. $-\frac{33}{2}$

D. $-\frac{5}{2}$

17

★ Mark for Review

ABC

A model estimates that from 2015 to 2020, the number of squirrels in a population increased by 150% of the previous year's value. If at the end of 2016, there were 180 squirrels, which equation represents this model, where n is the estimated number of squirrels t years after 2015 and $t \leq 5$?

A. $n = 72(1.5)^t$

B. $n = 72(2.5)^t$

C. $n = 180(1.5)^t$

D. $n = 180(2.5)^t$

18

★ Mark for Review

ABC

The function $f(x) = |59 - 2x|$ is defined. For which of the following values of k does $f(k) = 3k$?

A. $\frac{59}{5}$

B. $\frac{59}{2}$

C. $\frac{177}{5}$

D. 59

19

★ Mark for Review

ABC

The population P of a certain city y years after the last census is modeled by the equation $P = P_0(1+r)^y$, where r is a constant and P_0 is the population when $y = 0$. If the population decreases each year by a fixed percent, which of the following must be true?

A. $r < -1$

B. $-1 < r < 0$

C. $0 < r < 1$

D. $r > 1$

20

★ Mark for Review

ABC

The function $f(x) = 5,470(0.64)^{\frac{x}{12}}$ gives the value, in dollars, of a certain piece of equipment after x months of use. If the value decreases each year by $p\%$, what is the value of p ?

A. 4

B. 5

C. 36

D. 64

21

★ Mark for Review

ABC

The function $f(x) = \frac{1}{9}(x - 7)^2 + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving, where $0 \leq x \leq 10$. What is the best interpretation of the vertex of the graph $y = f(x)$?

A. The metal ball's minimum height was 3 inches above the ground.

B. The metal ball's minimum height was 7 inches above the ground.

C. The metal ball's height was 3 inches above the ground when it started moving.

D. The metal ball's height was 7 inches above the ground when it started moving.

22

★ Mark for Review

ABC

The function $f(t) = 8,000(0.65)^t$ models the number of coupons a company sent to customers at the end of each year, where t is the number of years since the end of 1998 and $0 \leq t \leq 5$. What is the best interpretation of the y -intercept of the graph of $y = f(t)$?

A. The minimum estimated number of coupons the company sent to their customers during the 5 years was 1,428.

B. The minimum estimated number of coupons the company sent to their customers during the 5 years was 8,000.

C. The estimated number of coupons the company sent to their customers at the end of 1998 was 1,428.

D. The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

23

★ Mark for Review

ABC

In the xy -plane, a parabola has vertex $(9, -14)$ and intersects the x -axis at two points. If the equation of the parabola is written as $y = ax^2 + bx + c$, what could be the value of $a + b + c$?

A. -23

B. -19

C. -14

D. -12

24

★ Mark for Review

ABC

The table below gives selected values of a polynomial function p :

x	$p(x)$
-2	5
-1	0
0	-3
1	-1
2	0

Based on the values in the table, which of the following must be a factor of p ?

A. $(x - 3)$

B. $(x + 3)$

C. $(x - 1)(x + 2)$

D. $(x + 1)(x - 2)$

25

★ Mark for Review

ABC

When the quadratic function f is graphed in the xy -plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the x -intercepts of this graph is $\left(-\frac{17}{4}, 0\right)$. What is the other x -intercept?

A. $\left(-\frac{29}{4}, 0\right)$

B. $\left(-\frac{7}{4}, 0\right)$

C. $\left(\frac{5}{4}, 0\right)$

D. $\left(\frac{17}{4}, 0\right)$

26

★ Mark for Review

ABC

The function $f(x) = (x + 3)(x + 1)$ is graphed in the xy -plane. Which of the following intervals contains the x -coordinate of the vertex of the graph of f ?

A. $-4 < x < -3$ B. $-3 < x < 1$ C. $1 < x < 3$ D. $3 < x < 4$ **27**

★ Mark for Review

ABC

A landscaper is designing a rectangular garden. The length is 5 feet longer than the width. If the area is 104 square feet, what will be the length in feet?

28

★ Mark for Review

ABC

$P(t) = 290(1.04)^{\frac{4}{6}t}$ models the population, in thousands, of a certain city t years after 2005. The population increases by $n\%$ every 18 months. What is the value of n ?

A. 0.38

B. 1.04

C. 4

D. 6

29

★ Mark for Review

ABC

For the function f , $f(0) = 86$, and for each increase in x by 1, $f(x)$ decreases by 80%. What is the value of $f(2)$?

30

★ Mark for Review

ABC

$M = 1,800(1.02)^t$ models gym members, M , of a gym t years after opening. Which equation models the number of members q quarter-years after opening?

A. $M = 1,800(1.02)^{q/4}$

B. $M = 1,800(1.02)^{4q}$

C. $M = 1,800(1.005)^{4q}$

D. $M = 1,800(1.082)^q$

31

★ Mark for Review

ABC

The functions f and g are defined by the given equation, where $x \geq 0$. Which function shows a maximum value as a constant or coefficient for $x \geq 0$?

I $f(x) = 33(0.4)^{x+3}$

II $g(x) = 33(0.16)(0.4)^{x-2}$.

A. I only

B. II only

C. I and II

D. Neither I nor II

32

★ Mark for Review

ABC

Square P has a side length of x inches. Square Q has a perimeter that is 176 inches greater than the perimeter of square P . The function f gives the area of square Q , in inches. Which of the following defines f ?

A. $f(x) = (x + 44)^2$

B. $f(x) = (x + 176)^2$

C. $f(x) = (176x + 44)^2$

D. $f(x) = (176x + 176)^2$

33

★ Mark for Review

ABC

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

A. $\frac{a}{4}$

B. a

C. $4a$

D. $6a$

34

★ Mark for Review

ABC

The function $h(x) = a^x + b$, where a and b are positive constants, passes through $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab ?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 54

D. 60

35

★ Mark for Review

ABC

Table:

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y . Given $y = f(x) + 4$, and f is quadratic function, what is the y -coordinate of the y -intercept of $y = f(x)$?

36

★ Mark for Review

ABC

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 10 feet above the ground and reaches its maximum height of 1,034 feet above the ground 8 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 10 seconds after being launched?

A. 234

B. 778

C. 970

D. 1,014

37

★ Mark for Review

ABC

Function f is defined by $f(x) = (x + 6)(x + 5)(x + 1)$. Function g is defined by $g(x) = f(x - 1)$. The graph of $y = g(x)$ in the xy -plane has x -intercepts at $(a, 0)$, $(b, 0)$, and $(c, 0)$, where a , b , and c are distinct constants. What is the value of $a + b + c$?

A. -15

B. -9

C. 11

D. 15

38

★ Mark for Review

ABC

The total distance d , in meters, traveled by an object moving in a straight line can be modeled by a quadratic function that is defined in terms of t , where t is the time in seconds. At a time of 10.0 seconds, the total distance traveled by the object is 50.0 meters, and at a time of 20.0 seconds, the total distance traveled by the object is 200.0 meters. If the object was at a distance of 0 meters when $t = 0$, then what is the total distance traveled, in meters, by the object after 30.0 seconds?

39

★ Mark for Review

ABC

$f(x) = 4x^2 - 50x + 126$ The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

40

★ Mark for Review

ABC

$$h(x) = x^3 + ax^2 + bx + c$$

The function h is defined above, where a , b , and c are integer constants. If the zeros of the function are -5 , 6 , and 7 , what is the value of c ?

41

★ Mark for Review

ABC

$$f(x) = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x+2)$, which of the following equations defines the function g ?

A. $g(x) = 18(4)^x$

B. $g(x) = 144(4)^x$

C. $g(x) = 18(8)^x$

D. $g(x) = 81(16)^x$

42

★ Mark for Review

ABC

$$f(x) = (x + 7)^2 + 4$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

43

★ Mark for Review

ABC

Kao measured the temperature of a cup of hot chocolate placed in a room with a constant temperature of 70 degrees Fahrenheit ($^{\circ}\text{F}$). The temperature of the hot chocolate was 185°F at 6:00 p.m. when it started cooling. The temperature of the hot chocolate was 156°F at 6:05 p.m. and 135°F at 6:10 p.m. The hot chocolate's temperature continued to decrease. Of the following functions, which best models the temperature $T(m)$, in degrees Fahrenheit, of Kao's hot chocolate m minutes after it started cooling?

A. $T(m) = 185(1.25)^m$

B. $T(m) = 185(0.85)^m$

C. $T(m) = (185 - 70)(0.75)^{\frac{m}{5}}$

D. $T(m) = 70 + 115(0.75)^{\frac{m}{5}}$

44

★ Mark for Review

ABC

$$p(t) = 90,000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

A. $r(m) = \frac{90,000}{12}(1.06)^m$

B. $r(m) = 90,000 \left(\frac{1.06}{12} \right)^m$

C. $r(m) = 90,000 \left(\frac{1.06}{12} \right)^{\frac{m}{12}}$

D. $r(m) = 90,000(1.06)^{\frac{m}{12}}$

45

★ Mark for Review

ABC

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years t it will take for the population of the town to reach 60,000?

A. $50,000 = 60,000(0.03)^t$

B. $50,000 = 60,000(3)^t$

C. $60,000 = 50,000(0.03)^t$

D. $60,000 = 50,000(1.03)^t$

46

★ Mark for Review

ABC

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A. $f(x) = 50(2)^{x+1}$

B. $f(x) = 80(2)^x$

C. $f(x) = 128(2)^{x-1}$

D. $f(x) = 205(2)^{x-2}$

47

★ Mark for Review

ABC

$$h(x) = -16x^2 + 100x + 10$$

The quadratic function above models the height above the ground h , in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy -plane, which of the following represents the real-life meaning of the positive x -intercept of the graph?

A. The initial height of the projectile

B. The maximum height of the projectile

C. The time at which the projectile reaches its maximum height

D. The time at which the projectile hits the ground

48

★ Mark for Review

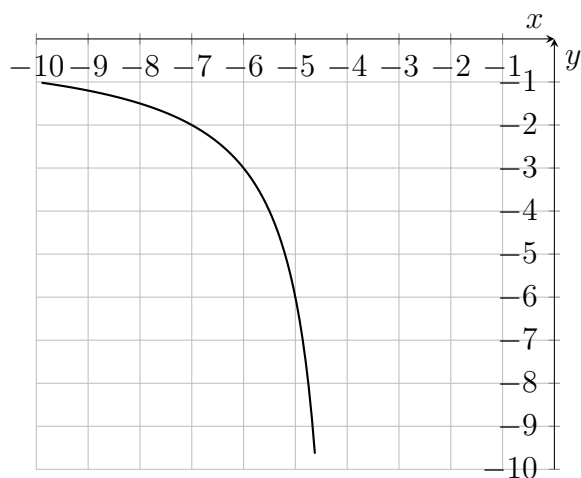
ABC

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function f , the table above shows several values of x and their corresponding values of $f(x)$, where a is a constant greater than 1. If k is a constant and $f(k) = a^{29}$, what is the value of k ?

49

★ Mark for Review

ABC

The rational function f is defined by an equation in the form $f(x) = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = f(x)$ is shown. If $g(x) = f(x+4)$, which equation could define function g ?

A. $g(x) = \frac{6}{x}$

B. $g(x) = \frac{6}{x+4}$

C. $g(x) = \frac{6}{x+8}$

D. $g(x) = \frac{6(x-4)}{x+4}$

50

★ Mark for Review

ABC

$$f(x) = (x - 10)(x + 13)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

A. -130

B. -13

C. $-\frac{23}{2}$

D. $-\frac{3}{2}$

51

★ Mark for Review

ABC

For the function q , the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

A. $q(x) = 0.55(14)^x$

B. $q(x) = 1.45(14)^x$

C. $q(x) = 14(0.55)^x$

D. $q(x) = 14(1.45)^x$

52

★ Mark for Review

ABC

Two variables, x and y , are related such that for each increase of 1 in the value of x , the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

A. $y = 4(x)^{200}$

B. $y = 4(200)^x$

C. $y = 200(4)^x$

D. $y = 200(4x)$

53

★ Mark for Review

ABC

What is the minimum value of the function f defined by $f(x) = (x - 2)^2 - 4$?

A. -4

B. -2

C. 2

D. 4

54

★ Mark for Review

ABC

Let the function p be defined as $p(x) = \frac{(x - c)^2 + 160}{2c}$, where c is a constant. If $p(c) = 10$, what is the value of $p(12)$?

A. 10.00**B.** 10.25**C.** 10.75**D.** 11.00